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Impact of Microparticle Orientation and Fractal Geometry on the Dynamic Response of Magnetorheological Elastomers.

J. Luis Color-Gomez[a] Luis M. Palacios-Pineda[a] Imperio A. Perales-Martínez[b] Alex Elias-Zúñiga[b]

[a] *Tecnológico Nacional de México / Instituto Tecnológico de Pachuca, Carr. México-Pachuca Km 87.5, 42080 Pachuca, Hgo., México*
[b] *Mechanical Engineering and Advanced Materials Department; Institute of Advanced Materials for Sustainable Manufacturing, Tecnologico de Monterrey, Ave. Eugenio Garza Sada 2501, C.P. 64849, Monterrey, N.L., Mexico.*

ABSTRACT

This study explores the influence of microparticle fractal alignment on the magneto-mechanical behavior of magnetorheological elastomers (MREs). By varying the orientation of carbonyl iron particle chains within a silicone matrix, the research demonstrates that the alignment angle plays a decisive role in tuning the material's mechanical stiffness and magnetorheological response. Results show that smaller angles between the magnetic field and particle chains yield higher stiffness and more pronounced field-induced effects. Interestingly, for certain orientations, increasing the filler content does not significantly enhance the storage modulus, highlighting the complex interplay between particle concentration and alignment. Moreover, the magnetorheological effect diminishes with increasing strain, regardless of alignment or particle content behavior attributed to particle network disruption. The study also establishes a correlation between the fractal dimension of particle arrangements and viscoelastic properties, enabling predictive modeling through the fractional Zener model. These findings contribute to a deeper understanding of MRE behavior and offer valuable insights for designing advanced smart materials for adaptive mechanical systems and magnetic field-responsive applications.

Introduction

- The growing interest in smart materials has led to extensive research on Magnetorheological Elastomers (MREs) due to their tunable mechanical properties under magnetic fields. This study explores MREs from a novel angle, focusing on how microparticle chain orientation and fractal properties influence their magneto-mechanical behavior. Different alignments were induced using magnetic fields, combined with high efficiency materials and varying filler concentrations during fabrication.
- Previous studies have highlighted the critical role of microparticle alignment angles in defining stiffness, storage modulus, and magneto-induced effects. Models based on dipole interactions and experiments under dynamic and static conditions (e.g., shear, compression) revealed that orientations between 0° and 65° significantly impact performance. Special arrangements like "Fountain-like" structures further demonstrated enhanced mechanical responses. Notably, the magnetorheological effect often weakens with increasing strain due to particle network disruption, regardless of alignment or concentration.
- Building on this, the present work investigates the correlation between fractal dimension (FD) of particle arrangements and viscoelastic behavior. Through rheological tests and FD calculation, a strong relationship was found, enabling dynamic property prediction via models like the Balankin equation. This study contributes to the design of advanced MREs by linking microstructural features with functional properties, offering insights for smart, field-responsive applications.

Materials and methods

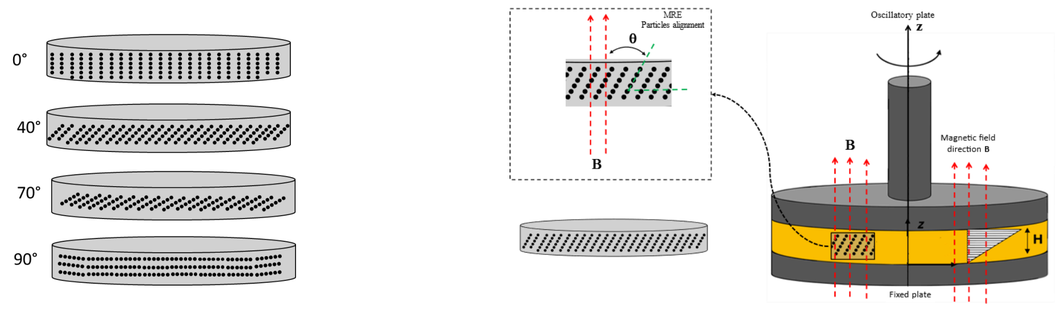
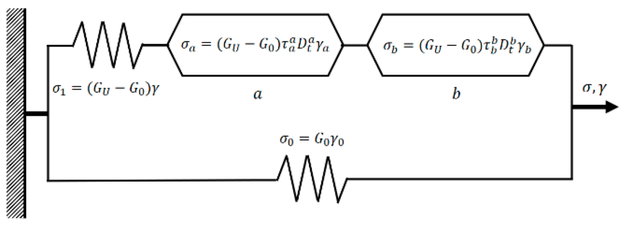
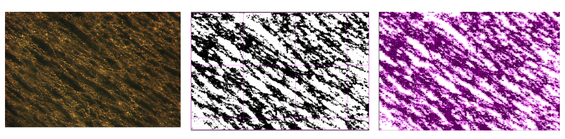


Figure 1. Definition of angles used in the present work, θ is the angle between the magnetic field and the particles alignment, used along the experiments and in the data processing [1].



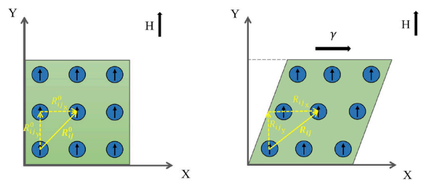
$$G^*(i\omega) = G' + iG'' = \frac{G_U + G_0[(i\omega\tau_a)^{-a} + (i\omega\tau_b)^{-b}]}{1 + (i\omega\tau_a)^{-a} + (i\omega\tau_b)^{-b}}$$

Figure 2. Fractional Zener model [2].



$$F_D = \frac{\log N_\delta(F)}{\log \frac{1}{\delta}} \qquad \zeta_r \sim a^{FD},\ \zeta_r \sim b^{FD} \qquad b = \sqrt[FD]{\zeta_r}$$

Figura 3. The box-counting method and the Fractal influence in the Zener model [3].



$$\Delta G = \frac{2\emptyset A\mu_0\mu_1\left(\frac{a}{r_0}\right)^3 k}{\gamma}\left[\frac{\chi H_0}{1 - 2C\chi\left(\frac{a}{r_0}\right)^3 \cos^5\theta + \frac{2}{3}C\chi\cos^3\theta}\right]^2$$

Figura 4. Magneto-induced shear modulus [4].

Results

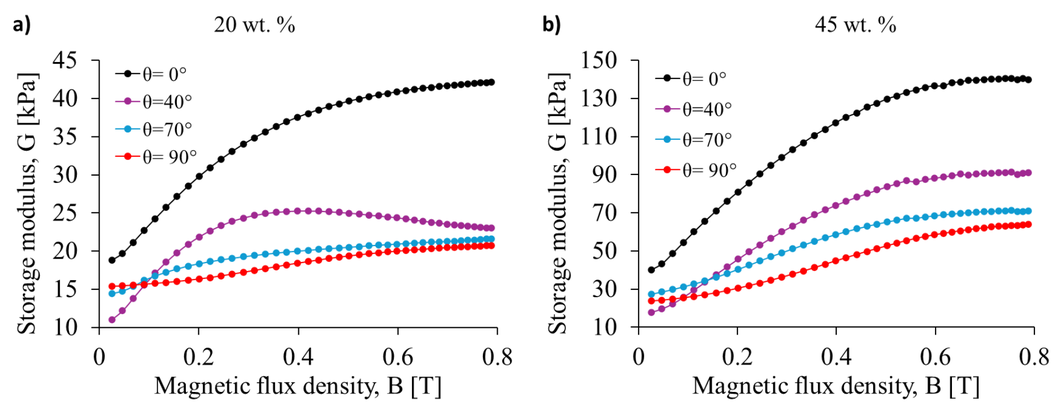


Figure 5. Storage modulus variation with respect to the magnetic field applied to the different particles' alignments inside the composite material, a) for the MRE with 20 wt.% of CIPs, b) for the MRE with 45 wt.% of CIPs. These rheological tests were performed at room temperature of 25 °C with a shear strain constant of , and an oscillatory frequency of , the applied normal force was kept constant at a value of F = 1.5N.

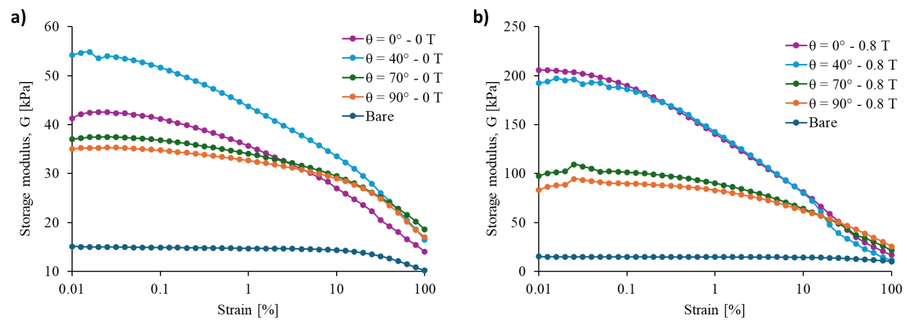


Figure 6. MREs magneto-induced modulus variation with respect to the strain applied to the different particles' alignments inside the composite material with a) 20 and b) 45 wt.% of CIPs. These rheological tests were performed at room temperature of 25 °C with a constant oscillatory frequency of , and an applied constant normal force of , and magnetic field of and .

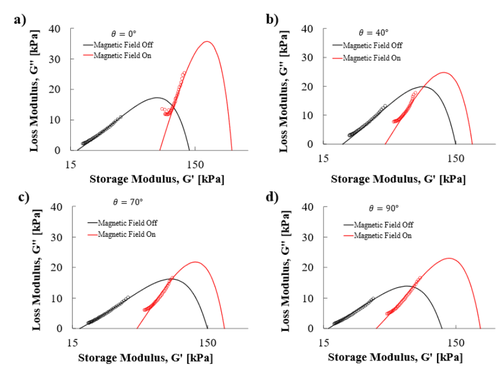


Figure 7. Cole-Cole diagram for 45 wt.% of CIPs when the magnetic field is off or On in four different angles a) , b) , c) , and d) .

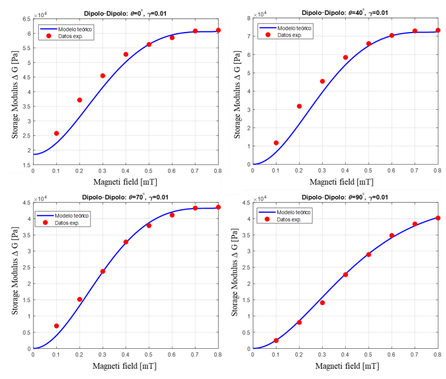


Figure 8. Comparation between Magneto-induced modulus model and the experimental results, The solid blue line represents the magnetic dipole model for an aggregate of 45 wt.% of CIPs..

Conclusions

This study highlights the critical role of microparticle alignment in tuning the magneto-mechanical behavior of magnetorheological elastomers (MREs). Smaller alignment angles relative to the magnetic field significantly increase stiffness and the magnetorheological effect, while larger angles reduce the impact of filler concentration. Strain-induced reductions in magneto-induced modulus point to particle network disruption under dynamic loading.

A key contribution is the correlation established between the fractal dimension of particle chains and the material's viscoelastic response, enabling predictive modeling via fractional viscoelastic frameworks. XRD and FTIR analyses confirm structural changes without chemical alteration, and thermogravimetric data show improved thermal stability with higher filler content.

By combining experimental analysis with dipole-based theoretical modeling, this research provides a comprehensive framework to understand and engineer the behavior of MREs, paving the way for smart materials with tunable properties in adaptive and multifunctional systems.

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Autor

Ph.D. student in Mechanical Engineering with a Master's degree in Mechanical Engineering. He is currently conducting a doctoral research stay at the Institute of Advanced Materials for Sustainable Manufacturing at Tecnológico de Monterrey and participates in international collaborative projects. His research focuses on smart materials, integrating magnetism, fractal architectures, and mechanics to develop magnetorheological elastomers (MREs) with programmable properties. He has published papers in high-impact journals and is actively exploring additive manufacturing techniques for the fabrication of functional components in soft robotics.